PDF rendering note: this is a reflowed export of the complete filled Word form, produced by the reviewer with ReportLab. The editable DOCX is the original-layout deliverable. Word COM export did not complete; this PDF does not claim exact Word pagination.

INDEPENDENT REVIEW PACKAGE

Dragon Market Transactional Correctness & Research Baseline Review

Reviewer instructions • hands-on test plan • evidence log • approval form

Target repository	github.com/houtanrocky/dragon-market
Exact reviewed revision	f4ba21185d1ba7b4bbb56f5addcc27d1b75bdec7
Review status	AI REVIEW: CHANGES REQUIRED. Independent human approval remains pending.
Pack prepared	18 October 2026 | Reviewer copy 1.0
Intended use	Independent assessment before MSc coding-agent experiments

IMPORTANT This is an editable checklist and sign-off document, not a claim that Dragon Market is bug-free. Mark every test with the result you personally observed; never approve based only on the coding agent's previous test report.

COMPLETED AI REVIEW — Codex executed this review on 2026-10-21. No independent human review or researcher approval is claimed. Existing initially clean exact-SHA checkout used at requester instruction. All 68 activity results are explicitly completed; optional research task checks are NOT RUN. Three findings remain open. Reviewer declaration and typed acknowledgement are in Section 11.

For invited reviewers

Please review only the areas within your expertise. Functional/UI reviewers can assess visible workflows; Go/PostgreSQL reviewers should evaluate invariants, SQL effects and concurrency; reproducibility reviewers can independently rebuild and inspect evidence. Record the scope of your review and sign Section 11.

You may fill this Word document electronically or print it. In a result cell, tick one status: P = PASS; F = FAIL; B = BLOCKED; NR = NOT RUN; N/A = NOT APPLICABLE. Do not leave required sections ambiguous.

Review outcomes

APPROVE: no unresolved material issues within the reviewed scope. CONDITIONAL: specific unresolved minor items are listed. REJECT / CHANGES REQUIRED: material correctness or reproducibility concerns. A reviewer's signature attests only to the checks they performed; the researcher makes the final baseline-freeze decision.

1. Assignment, safety and evidence rules

Reviewer details — complete before testing

Reviewer name	Codex (OpenAI AI coding assistant)
Contact / affiliation	AI reviewer in this session; no personal contact address or institutional appointment. Does not sign on behalf of OpenAI.
Review date (UTC)	2026-10-21 UTC
Role(s)	☒ UI / functional ☒ Go/backend ☒ PostgreSQL ☒ Reproducibility ☒ Accessibility (limited automated/keyboard scope)
Testing scope	☐ Local clean checkout ☒ Existing local install ☐ Hosted review URL ☐ Source-only. Exact SHA initially clean; disposable DB harness.
Independent of author?	☐ Yes ☒ No / not established: AI reviewer requested by owner; not an independent human. No implementation authored in this review; earlier authorship independence cannot be certified.

Rules

• Never use a production database, real financial information, or shared user records. Use the local dev database or the isolated test harness.

• Do not use docker compose down -v, reset unknown volumes, replay SQL migrations on a populated database, or bypass migration-11 safeguards.

• Do not push fixes directly while reviewing. Report a finding with an exact reproduction; proposed patches should use separate branches.

• Do not share passwords, database URLs, bearer tokens, full environment dumps, or private test oracles in screenshots or reports.

• For every failure, preserve the exact SHA, command/request, expected rule, actual observation, logs or sanitized screenshot, and issue ID.

• Make no approval claim for tests you did not execute. Automated PASS is useful evidence but not proof of the rule's correctness.

• The web UI includes a Development Guild Switcher, not authentication; it is for local/demo testing only.

SCOPE CONTROL The primary manual-review application is the local workbench. A hosted review site is optional and may run a different SHA. If using one, record its displayed deployed SHA and do not treat it as a review of f4ba211 unless they match.

Evidence naming convention

Use e.g. R2-WAL-04-2026-10-18.txt, R1-UI-03.png, or a GitHub issue link. Keep evidence in a private folder until scrubbed. Document the Git SHA and UTC timestamp with each artifact.

2. Clean checkout and first-time setup

Prerequisites

Install Git, Go 1.25 or newer, Docker Desktop/Engine with Linux containers running, Node.js 24 LTS and npm. For Go race tests on Windows, a supported C compiler/CGO toolchain is required. Playwright/Chromium is needed for browser E2E tests.

Check installed tools (PowerShell or Bash)

```
git --version
go version
node --version
npm --version
docker version
docker compose version
```

Get the exact tested revision

Run in a NEW folder (not inside your existing working copy)

```
git clone https://github.com/houtanrocky/dragon-market.git
cd dragon-market
git switch --detach f4ba21185d1ba7b4bbb56f5addcc27d1b75bdec7
git rev-parse HEAD
git status --short
```

The SHA printed by git rev-parse HEAD must exactly equal f4ba21185d1ba7b4bbb56f5addcc27d1b75bdec7, and git status --short should be empty before testing. Detached HEAD is intentional for review; it prevents accidental branch updates.

3. Start the application and run automated checks

Launch the interactive workbench

From the cloned dragon-market directory

```
npm --prefix web ci
node scripts/workbench.mjs
```

Open http://127.0.0.1:5173 in your browser. The documented workbench uses PostgreSQL on 15432, Go API 8080, mock oracle 8090, optional lab runner 8082, and Vite 5173. These ports must be available. Keep the command running while performing browser checks.

On a fresh initialized database, seed guilds are guild-seller, guild-buyer, guild-bidder. Their raw balances are 10,000 integer units (= 100.00 display G), and daily limits are 5,000 units (= 50.00 G). Do not confuse raw integer amounts with displayed G.

```
FRESH DATABASE The workbench command does not guarantee an unused Docker volume if this computer previously ran Dragon Market. Confirm that migration 11 is present and the database is dedicated to review. The safest independent database exercises use Testcontainers/E2E, not deletion of an unknown Compose volume.
```

Baseline automated verification — record each outcome

Run with Docker engine active; save raw outputs

```
go build ./...
go test -count=1 ./...
go test -race -count=1 ./...
go vet ./...
node scripts/verify.mjs
node scripts/mutation-check.mjs
```

Focused wallet regression and fault-injection evidence

```
go test -v -count=1 ./internal/guild ./internal/postgres -run '^TestDailyLimit'
node scripts/verify-transactional-baseline.mjs
node scripts/check-daily-limit-mutations.mjs
```

Browser end-to-end checks (isolated disposable DB)

```
npm --prefix web exec -- playwright install chromium
npm --prefix web run test:e2e
```

If Chromium installation is blocked and Chrome is installed, consult the README for PLAYWRIGHT_CHANNEL=chrome. Browser E2E uses disposable test containers and different ports; do not point it at the shared workbench database.

The focused verifier writes into its evidence folder; copy existing evidence elsewhere before rerunning. Mutation checks operate on disposable source copies, not the reviewed code. Record all FAILED, BLOCKED and NOT RUN statuses honestly.

3A. Automated execution log

Tick a status only after you personally observe the command output. Add evidence filename or issue ID in the right cell.

REVIEW ACTIVITY & ACCEPTANCE EVIDENCE	REVIEWER RESULT
AUTO-01 Git SHA and clean tree Expected / inspect: Printed exact target SHA; review checkout initially clean.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E01 Exact target SHA; initially clean existing checkout. No clone/pull; requester-authorized deviation.
AUTO-02 Go build and vet Expected / inspect: Build and vet complete with exit code 0.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E03 Original Go build and vet exit 0.
AUTO-03 Uncached Go tests Expected / inspect: All relevant unit, integration, and repository tests exit 0; no Testcontainers failures hidden.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E03 Uncached verbose suite exit 0; real PostgreSQL Testcontainers; no skipped tests observed.
AUTO-04 Race-enabled suite Expected / inspect: Race detector exits 0; record Windows CGO failures separately as BLOCKED.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E03 Windows CGO race suite exit 0; GCC 16.1.0 available.
AUTO-05 Full verify.mjs Expected / inspect: Complete verification reports successful backend and frontend checks; save results.json.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E04 Full verifier exit 0: build/vet/tests/race plus frontend typecheck/lint/15 tests/build.
AUTO-06 Baseline wallet regression Expected / inspect: Daily-limit targeted tests and transactional verifier pass with real database.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E05, E12 Focused daily-limit tests and final baseline verifier pass. First verifier attempt failed due to reviewer file packaging; preserved and corrected.
AUTO-07 Fault mutation checks Expected / inspect: All deliberate daily-limit wallet faults are detected as business assertion failures, not compile failures.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E06 All three daily-limit mutants rejected by named SQL-backed assertion; two purchase mutants also detected.
AUTO-08 Playwright end-to-end Expected / inspect: Real browser suite passes on a disposable database; collect trace/screenshot report.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E07 All 7 existing browser tests passed with installed Chrome and isolated disposable DB.

4. Manual browser review — functional testers

Choose a unique suffix for entities (e.g., R1-1810) to avoid mixing your results with another reviewer. Use the seeded guilds. Capture initial wallet and ownership values before changing anything. Perform UI tasks through real endpoints, not screenshots from a previous run.

4A. UI, market and accessibility

REVIEW ACTIVITY & ACCEPTANCE EVIDENCE	REVIEWER RESULT
UI-01 Start and health Expected / inspect: Workbench loads; System Status clearly displays actual environment, backend reachability and errors.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E08 System Status shows development, exact SHA, PostgreSQL identity, successful read and HTTP health.
UI-02 Identity selector Expected / inspect: Switch between guild-seller, guild-buyer and guild-bidder; wallet changes to selected guild. No claim of true authentication.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E08 Switched all three seeded guilds; initial visible 100.00 G; explicitly not authentication.
UI-03 Create item Expected / inspect: Create a common/rare item under seller; it appears with correct ID, owner, item type and price.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E08 Created uniquely named rare item; owner/type/700-unit price verified in real snapshot.
UI-04 Create fixed-price order Expected / inspect: List an eligible common/rare item; order becomes available to buyer. Invalid items/prices are rejected.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E08 Listed for 700; zero price blocked by native range validation; invalid API price rejected.
UI-05 Purchase and ownership Expected / inspect: Buy as another guild; final owner changes once; balances update consistently and order cannot be re-sold.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E08 Buyer 10000→9300, seller 10000→10700; owner transferred once; sold rebuy rejected.
UI-06 Cancel order Expected / inspect: An eligible order cancels; a bought or otherwise ineligible order cannot be cancelled incorrectly.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E08 Common-item listing cancelled to free; sold listing cancellation rejected (400).
UI-07 Create auction Expected / inspect: Create a legendary item auction; verify the displayed base price, deadline and starting state.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E08 Legendary 700-unit base item; active auction, backend deadline approximately 24 hours ahead.
UI-08 Bid and wallet reservations Expected / inspect: Bid as buyer/bidder; accepted bid changes reserved and available balance; total gold stays unchanged.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E08 500 bid reserves 500; total gold unchanged; available gold decreases by hold.
UI-09 Outbid/cancel behavior Expected / inspect: Place a higher valid bid; old bidder hold releases; cancel only eligible active highest bid.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E08 Raise 500→525 uses delta 25; bidder 552 outbids; old hold released; highest cancellation succeeds; repeated cancellation 409.
UI-10 Request replay Expected / inspect: Repeat an eligible financial request with the same Idempotency-Key; inspect persisted effect, not just response text.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E08 Same-key replay: 204 plus replay header; unchanged ownership/balances; exactly 2 audit rows and 1 idempotency record.
UI-11 Transaction Inspector Expected / inspect: Compare pre/post snapshots and IDs. Identify which values are raw persisted fields vs current-day projection.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E08, E09, E11 Inspected raw persisted pre/post values and date; distinguished them from GET-wallet projection. DM-REV-CODEX-001 documents wrong Wallet display.
UI-12 Errors and validation Expected / inspect: Try invalid amount, malformed data or forbidden transition. UI shows clear backend error without a misleading success message.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E08 526 rejected below required 552 with visible error; zero base price HTTP 400; no false success.

REVIEW ACTIVITY & ACCEPTANCE EVIDENCE	REVIEWER RESULT
UI-13 Accessibility/responsiveness Expected / inspect: Keyboard Tab/focus, visible labels, appropriate contrast, mobile viewport, refresh and navigation work.	☐ P ☒ F ☐ B ☐ NR ☐ N/A Evidence: E10; DM-REV-CODEX-003 Mobile audit scroller lacks keyboard focus (axe serious). Desktop scan, skip link, route refresh and page-width checks passed.
UI-14 Safety boundaries Expected / inspect: Research Lab labels C/F/G local-only where applicable; no unrestricted remote reset, SQL, or hidden test routes.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E07, E08, E11 Local harness executes C/F/G in isolated DBs; hosted gateway tests reject harness/arbitrary debug routes. No live hosted site assessed.

4B. Record a complete purchase example

Item and order IDs	Item: bd3c9cf4-3acc-40c7-b64f-842c3e02ca84 Order: 5f1a6680-a565-410b-af1f-0659d893342f Codex-R1-2110-Emberglass; raw price 700 = 7.00 G.
Seller / buyer guilds	guild-seller → guild-buyer; isolated dragon_research_disposable database.
Initial wallet snapshots	Before purchase: seller gold=10000/reserved=0/spent=0; buyer gold=10000/reserved=0/spent=0; available equals gold. Limit=5000 each.
Final wallet snapshots	After purchase and same-key replay: seller gold=10700/reserved=0/spent=0; buyer gold=9300/reserved=0/spent=700; audit rows=2; idempotency rows=1. All amounts raw units.
Ownership before / after	guild-seller → guild-buyer once; order sold; same-key replay retains buyer; new-key rebuy rejected.
Same-key replay outcome	204 and Idempotency-Replayed:true; no new monetary, ownership or ledger effects (independent SQL).
Screenshot / API evidence	E08: browser/purchase-example.json; browser/observations.json; browser/purchase-replay-inspector.png; browser/browser-trace.zip.

5. Business invariant correctness — expert code review

CRITICAL RULE The key rule selected by the project is: S(d) + R <= L, where S(d) is completed spending on the current UTC day, R is all outstanding holds regardless of age, and L is the positive daily limit. A zero limit means unlimited. The accounting rule is a specification to critically assess, not something reviewers must automatically agree with.

Wallet transition expectations

Operation	Completed spend S	Reserved R
Reserve a	unchanged	+a
Release a	unchanged	−a
Spend a	+a	unchanged
Settle/Deduct a	+a	−a
Earn a	unchanged	unchanged
UTC date rollover	reset to 0	unchanged

Also verify: total gold changes only for actual debits/earnings; available gold = gold − reserved; reserves and spend require adequate capacity; settlement converts an existing hold to spending without double-counting.

5A. Wallet semantics and edge cases

REVIEW ACTIVITY & ACCEPTANCE EVIDENCE	REVIEWER RESULT
WAL-01 Policy acceptance Expected / inspect: Independently assess whether completed UTC-day spending plus every outstanding historical reservation is the intended limit rule.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E05, E11 Supplied S(current UTC day)+all holds≤positive L is coherent under valid starting state, fixed limits and reliable UTC clock; L=0 unlimited. Human intent confirmation remains owner-owned.
WAL-02 Reservation capacity Expected / inspect: New hold accepted only if funds and S + R + new hold fit positive L; rejection has zero persisted changes.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E03, E05 Funds/capacity checks and row locks enforce reserve allowance; failed operations have no committed effect.
WAL-03 Release after old hold Expected / inspect: Release reduces R without changing already completed S; yesterday's hold release must never erase today's spending.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E05, E06 Old hold release reduces reserved only; mixed-age test and mutation rejection preserve completed spending.
WAL-04 Old hold at UTC rollover Expected / inspect: At midnight UTC S resets; existing R stays. A hold of 500 with limit 500 leaves no capacity for another spend of 300 before release.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E05 500 hold at 500 limit rejects 300 spend before release; Tehran midnight does not reset UTC allowance.
WAL-05 Spend 300 then reject 500 Expected / inspect: After the above hold is released and 300 is spent today, a further 500 must fail under limit 500; only 300 debit persists.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E05 After release, 300 debit persists; next 500 rejected; SQL gold=9700/reserved=0/completed=300/audits=2.
WAL-06 Settle old reserved amount Expected / inspect: Converting reserved a to completed spending a preserves S+R and changes total gold exactly once.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E05 Old settlement exchanges reserved for spending; SQL gold debited once; repeated settlement rejected.
WAL-07 Cancel same-day bid Expected / inspect: Cancellation frees reservation headroom without refunding completed debits from same day.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E05, E08 Cancellation/outbid restores reservation capacity while completed 700-unit purchase remains charged.
WAL-08 Earnings do not reset allowance Expected / inspect: An earn increases gold but does not reduce S(d) or restore daily spending allowance.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E09 Independent SQL test: spend 500, earn 1000, next spend 1 rejected; gold=10500/spent=500/audits=2.
WAL-09 Exact limit and zero limit Expected / inspect: S+R exactly L is accepted where otherwise valid; > L rejected. L=0 uses documented unlimited rule.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E03, E05 Exact boundary accepted; over-cap rejected; zero-limit behavior and integer limits exercised.
WAL-10 Integer bounds and invalid amounts Expected / inspect: Negative, zero, insufficient funds/reserves and integer overflow cannot corrupt stored accounting.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E03, E05 Positive amounts, insufficient funds/reserve and int64 overflow checked in domain suites.
WAL-11 Projection and raw audit fields Expected / inspect: GET wallet projects current UTC spend without mutating history; diagnostic raw daily_spent must be interpreted with daily_spent_on.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E05, E09, E11 GET wallet projects prior-day counter to 0 without write; raw inspector retains 300 and prior date. UI defect separately reported.
WAL-12 Independently inspect ledger Expected / inspect: Compare transaction/audit rows and wallet state with the expected arithmetic; do not rely only on UI or service response.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E08, E09 Independently queried PostgreSQL wallet/item/audit rows and debit totals; no reliance on UI responses alone.

6. Transaction, order and auction correctness

6A. Financial and concurrency semantics

REVIEW ACTIVITY & ACCEPTANCE EVIDENCE	REVIEWER RESULT
TX-01 Double-purchase prevention Expected / inspect: Two concurrent buyers cannot both purchase one listing; only one owner and intended persisted money effects remain.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E03, E07 Concurrent-buy and Research Lab tests: single winner, intended owner and committed ledger effects.
TX-02 Purchase atomicity Expected / inspect: If purchase fails during transaction, ownership, balances and relevant audit records all roll back.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E03, E07 Controlled late failure after debit/earn/transfer rolls back every persisted effect and idempotency claim.
TX-03 Same idempotency key Expected / inspect: Replaying completed buy/bid returns original logical result; no duplicated debit, hold, item transfer or ledger rows.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E03, E08 Buy/bid replay suites inspect SQL; UI exact purchase replay produces no extra ledger entries.
TX-04 Different keys/new operation Expected / inspect: New key means a distinct attempt and must still enforce item/order and wallet validity.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E03, E08 New-key attempt on sold order rejected; changed same-key payload covered by existing SQL-backed test.
TX-05 Auction unique activation Expected / inspect: A legendary item cannot have two simultaneously active auctions when the rule forbids it.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E03 Concurrent auction starts yield one active auction; partial unique index and item transition reviewed.
TX-06 Bid minimum increment Expected / inspect: Bids below required 5% step are rejected; acceptable amounts respect integer rounding defined by implementation.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E03, E08 ceil(current/20) minimum increment reviewed; 525→552 boundary and 526 rejection observed; overflow test passes.
TX-07 Outbid release atomicity Expected / inspect: Old bidder reservation releases and new bidder reservation applies in same consistent transaction.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E03, E08 Outbid release and new hold within outer transaction; concurrent-bid SQL test leaves one active highest reservation.
TX-08 Same bidder raises bid Expected / inspect: Delta amount is reserved correctly; no duplicate full reservation.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E05, E08 Same-bidder delta reserve confirmed: total reservation 525, not 1025.
TX-09 Highest-bid cancellation Expected / inspect: Only active highest bid can be cancelled where allowed; release affects correct wallet and bid state.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E03, E08 Only eligible active highest cancellation releases correct wallet; repeated invalid cancellation 409.
TX-10 Deadline extension Expected / inspect: A valid bid in final five minutes extends deadline by five minutes under documented rules.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E03, E11 Controlled-clock auction test verifies final-five-minute extension; default config is 5-minute window/extension.
TX-11 Settlement idempotence Expected / inspect: Simultaneous/repeated settlement cannot duplicate transfer or debit; winning hold converts to spend exactly once.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E03, E05, E07 Worker plus concurrent settlement driver and repeat batch produce one debit/credit/ownership transfer.
TX-12 Concurrent reserve/spend Expected / inspect: Competing operations on one wallet cannot collectively exceed its allowed capacity.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E05 12 competing reserve/spend operations accept exactly 5×100 under limit 500; SQL usage=500.

REVIEW ACTIVITY & ACCEPTANCE EVIDENCE	REVIEWER RESULT
TX-13 Audit insertion rollback Expected / inspect: Force a controlled audit-write failure in isolated tests; date/accounting/ownership changes must roll back.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E03, E05 Injected audit failure rolls back accounting and stored day; purchase late-failure test rolls back ownership.
TX-14 Locking and nested transaction use Expected / inspect: Read SQL paths: FOR UPDATE and reuse of outer transaction protect correctness; no lock-order hazards introduced.	☐ P ☒ F ☐ B ☐ NR ☐ N/A Evidence: E09, E11; DM-REV-CODEX-002 Nested transactions reuse outer transaction, but buyer-then-seller wallet locks deadlock reciprocal purchases (40P01); one rolls back. No automatic retry.

Required expert commentary

Transaction boundary / locking issues observed
DM-REV-CODEX-002: nested transaction reuse and rollback work, but opposite buyer/seller lock order causes 40P01 under reciprocal purchases. E09 independent-tests.log and E11 source-excerpts.txt. No financial conservation failure observed.

Invariant counterexamples or missing concurrency interleavings
One naturally possible reciprocal-purchase schedule aborts one valid transaction. Existing same-item competition tests do not cover distinct items sharing two wallets in opposite directions. Cross-auction reciprocal wallets and worker starvation remain untested concerns, not claimed reproductions.

7. Schema, migration and reproducibility safety

7A. PostgreSQL and environment

REVIEW ACTIVITY & ACCEPTANCE EVIDENCE	REVIEWER RESULT
DB-01 New database migrations Expected / inspect: Fresh PostgreSQL receives migrations 1–11 in order and seed after migrations, and starts successfully.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E03, E07, E08 Fresh Testcontainers/E2E apply migrations 1–11 before seeding; v2 wallets and successful API startup observed.
DB-02 Legacy populated schema refusal Expected / inspect: Migration 11 refuses populated legacy guild data rather than silently reinterpreting mixed counters; original data unchanged.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E03, E05 Legacy initialization and focused migration tests reject populated v1 without changed balances/counters/ledger.
DB-03 Unsafe downgrade refusal Expected / inspect: Down migration refuses populated wallets; no silent data loss or inconsistent legacy rollback.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E09 Reviewer ran down migration on populated v2; explicit refusal; marker=2/gold=9877/spent=123 preserved.
DB-04 New code against v1 schema Expected / inspect: Wallet service fails clearly if daily_accounting_version=2 schema is absent; no silent fallback.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E09 Reviewer temp v1 schema: new wallet read rejects absent daily_accounting_version (42703); no silent fallback.
DB-05 Historical evidence preserved Expected / inspect: Original defect logs and prior candidate reports remain accessible; newer PASS does not rewrite historical FAIL.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E01, E14 All 234 tracked files byte-identical at completion, including historical FAIL/PASS evidence; temporary verifier output separately captured and restored.
DB-06 No destructive test cleanup Expected / inspect: Automated test harness only deletes its positively identified disposable containers; Compose data remains intact.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E14 Only positively identified disposable containers removed; no Compose launch/reset/down -v; zero running containers before and after.
DB-07 Local UTC dates Expected / inspect: Test local midnight that is not UTC midnight and UTC day rollover; recorded daily_spent_on matches chosen model.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E03, E05 Tehran midnight vs UTC midnight and persisted UTC day covered; carried reservations retained.
DB-08 Price-oracle fallback Expected / inspect: Bad/zero/negative/malformed/timeout oracle responses preserve last known good value; fixed order price stays fixed.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E13 Reviewer real HTTP oracle cases valid/zero/negative/malformed/503/timeout preserve cached 250; SQL base=250, fixed listing stays 100.
DB-09 Go/frontend separability Expected / inspect: Go backend can build/test without Node and UI does not implement separate monetary business logic.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E03, E11 Go build/test commands run independently of frontend tooling. Frontend formats/validates/suggests bids; server remains monetary authority.
DB-10 Reproduction evidence Expected / inspect: Log OS, Go/PostgreSQL/Node/Docker versions, test commands, SHA, exits, timestamps and any discrepancy.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E01, E02, E08 OS/tool/browser/PostgreSQL versions, SHA, timestamps, argv, exits, source hashes and deviations recorded.
DB-11 Review deployment identity (if used) Expected / inspect: Actual displayed deployed SHA and review DB name recorded; do not assume hosted URL runs f4ba211.	☐ P ☐ F ☐ B ☐ NR ☒ N/A Evidence: E01, E08 Hosted deployment not used. Local stack independently reports exact target SHA; no claims about Render/Neon/live CI.

REVIEW ACTIVITY & ACCEPTANCE EVIDENCE	REVIEWER RESULT
DB-12 Scope of security controls Expected / inspect: Guild switcher is not authentication; debug routes and Research Lab prohibited on untrusted public instances.	☒ P ☐ F ☐ B ☐ NR ☐ N/A Evidence: E03, E11 Local-only dev snapshot and harness restrictions plus gateway gate/origin/path tests reviewed. No production security certification.

Important migration warning

Migration 11 is deliberately a fresh-data boundary. A populated older database may already contain irrecoverable ambiguity in its old spending/reservation counter. Reviewers must NOT “fix” tests by clearing or transforming such data. Flag it for a separate reviewed reconciliation.

8. Independent review of research task packages

This optional-but-recommended specialist section concerns the separate agent-invariant-preservation repository. Baseline approval and task/oracle approval are TWO DIFFERENT decisions. The research manifest currently distinguishes operational target f4ba211 from task starting commit 58913f2; changes to either must be justified and evaluated independently.

Open a SEPARATE checkout for the research repository

```
git clone https://github.com/houtanrocky/agent-invariant-preservation.git
cd agent-invariant-preservation
git rev-parse HEAD
```

Do not place hidden reference patches or private evaluation tests inside the workspace later supplied to LLM agents. For private task/oracle review, the study owner should provide the exact task package and controlled access to references. A public repository alone may not include confidential oracles.

8A. T001 and T002 review

REVIEW ACTIVITY & ACCEPTANCE EVIDENCE	REVIEWER RESULT
TASK-01 T001 identity Expected / inspect: Record exact starting SHA and task ID: retry-safe purchase service outcome. Confirm requirement not already implemented as specified.	☐ P ☐ F ☐ B ☒ NR ☐ N/A Evidence: E11; scope exclusion Separate research repository, exact task packages, reference patches and controlled hidden oracles not supplied/reviewed. No task/oracle/fairness/leakage approval claimed.
TASK-02 T001 human reference Expected / inspect: Independent reviewer checks that reference patch implements requested behavior and preserves pre-existing purchase, wallet, audit and ownership rules.	☐ P ☐ F ☐ B ☒ NR ☐ N/A Evidence: E11; scope exclusion Separate research repository, exact task packages, reference patches and controlled hidden oracles not supplied/reviewed. No task/oracle/fairness/leakage approval claimed.
TASK-03 T001 hidden oracle Expected / inspect: Confirm baseline fails new feature test for the intended reason, reference passes, and faulty implementation is rejected.	☐ P ☐ F ☐ B ☒ NR ☐ N/A Evidence: E11; scope exclusion Separate research repository, exact task packages, reference patches and controlled hidden oracles not supplied/reviewed. No task/oracle/fairness/leakage approval claimed.
TASK-04 T002 identity Expected / inspect: Record exact starting SHA and task ID: retry-safe bid cancellation. Confirm precise semantics and allowed cancellation state.	☐ P ☐ F ☐ B ☒ NR ☐ N/A Evidence: E11; scope exclusion Separate research repository, exact task packages, reference patches and controlled hidden oracles not supplied/reviewed. No task/oracle/fairness/leakage approval claimed.
TASK-05 T002 human reference Expected / inspect: Review bid cancellation reference for reservation release, concurrency, idempotency and audit behavior.	☐ P ☐ F ☐ B ☒ NR ☐ N/A Evidence: E11; scope exclusion Separate research repository, exact task packages, reference patches and controlled hidden oracles not supplied/reviewed. No task/oracle/fairness/leakage approval claimed.
TASK-06 T002 hidden oracle Expected / inspect: Confirm baseline/reference/faulty-variant behavior as in TASK-03; test is not coupled to implementation details.	☐ P ☐ F ☐ B ☒ NR ☐ N/A Evidence: E11; scope exclusion Separate research repository, exact task packages, reference patches and controlled hidden oracles not supplied/reviewed. No task/oracle/fairness/leakage approval claimed.
TASK-07 Prompt fairness Expected / inspect: A and B have same underlying requirements, repo snapshot, tools and budgets; only presentation of invariant guidance differs.	☐ P ☐ F ☐ B ☒ NR ☐ N/A Evidence: E11; scope exclusion Separate research repository, exact task packages, reference patches and controlled hidden oracles not supplied/reviewed. No task/oracle/fairness/leakage approval claimed.
TASK-08 Leakage and provenance Expected / inspect: Tests/references excluded from agent workspace; source hashes, SHA, versions, evidence, and reviewer findings recorded.	☐ P ☐ F ☐ B ☒ NR ☐ N/A Evidence: E11; scope exclusion Separate research repository, exact task packages, reference patches and controlled hidden oracles not supplied/reviewed. No task/oracle/fairness/leakage approval claimed.

9. Evidence, exception and issue reporting

Record PASS only with inspectable evidence: command output, committed test, SQL snapshot, trace/screenshot, or code-review note with file:line. A reported bug that reproduces is useful research evidence; never change the expected result to make it disappear.

9A. Reviewer execution record

Review checkout SHA	f4ba21185d1ba7b4bbb56f5addcc27d1b75bdec7
Git status before tests	☒ Clean initially ☐ Not clean. After review: additive review/ only; all 234 tracked hashes unchanged.
OS / version	Windows 11 10.0.26200, amd64; Linux PostgreSQL via Docker Desktop.
Go / Node / npm	Go 1.26.0 windows/amd64; Node v24.18.0; npm 11.16.0; GCC 16.1.0; CGO race passed.
PostgreSQL / Docker	PostgreSQL 16.14 (postgres:16-alpine); Docker Desktop 4.58.0, Engine 29.1.5, Compose 5.0.1; Chrome 155.0.8059.85.
Database used	☒ Isolated Testcontainers ☒ Dedicated local disposable E2E DB ☐ Hosted review. No Compose/shared DB touched.
Workbench URL / SHA	http://127.0.0.1:5175 — disposable E2E workbench; API 8085, lab 8087. Actual displayed backend SHA: f4ba21185d1ba7b4bbb56f5addcc27d1b75bdec7. Shared 5173 launch intentionally replaced with isolated harness.
UTC start / end	2026-10-21T14:33:46.786657+00:00 → 2026-10-21T14:56:36+00:00 (UTC; testing completed before document sign-off).
Raw log folder / link	review/codex-2026-10-21/evidence/; E01–E14 index and SHA256SUMS.txt; packaged evidence ZIP.

9B. Finding template — duplicate this for each issue

Finding ID	DM-REV-CODEX-002
Severity	☐ Critical ☐ High ☒ Medium ☐ Low ☐ Observation
Failed test ID(s)	TX-14
Affected revision	f4ba21185d1ba7b4bbb56f5addcc27d1b75bdec7
How to reproduce	Run tools/run_independent.py against this exact checkout. The retained source copy adds only reviewer_independent_test.go. Create a rare listing owned by seller and another owned by buyer. Launch buyer buying seller's item and seller buying buyer's item concurrently. A test-only wallet wrapper waits after each real Spend has locked its buyer row; then lets both real Earn calls proceed. This deterministically schedules a naturally possible interleaving without changing application SQL or domain code.
Expected behavior	Valid reciprocal operations complete through consistent multi-wallet lock order or a bounded transaction retry; failed attempts must roll back.
Observed behavior	TestReviewerReciprocalPurchasesDoNotDeadlock fails: one service operation returns ERROR: deadlock detected (SQLSTATE 40P01), the other succeeds. SQL total gold=30000, audit rows=2, sold orders=1. Rollback/money conservation hold. HTTP 500 follows from handler mapping by source inspection; HTTP deadlock response was not separately executed.
Code / DB evidence	internal/order/service.go:132, :135; internal/postgres/guild_repo.go:60; internal/postgres/tx.go:37; internal/api/order_handler.go:188
Sanitized screenshot/log	E09 independent-tests.log; tools/independent_postgres_test.go.txt; E11 source-excerpts.txt
Suggested action	Acquire all participating wallet locks in a deterministic order, or deliberately retry the entire transaction on 40P01/40001 with bounded attempts. Retain rollback/idempotency guarantees. Review cross-auction two-wallet paths too.
Re-test status	☒ Open ☐ Fixed + verified ☐ Accepted risk ☐ Cannot reproduce

Detailed reproduction and SQL/state evidence
TestReviewerReciprocalPurchasesDoNotDeadlock fails: one service operation returns ERROR: deadlock detected (SQLSTATE 40P01), the

other succeeds. SQL total gold=30000, audit rows=2, sold orders=1. Rollback/money conservation hold. HTTP 500 follows from handler mapping by source inspection; HTTP deadlock response was not separately executed.
All three findings, exact reproductions and evidence links are completed in Appendix C and REVIEW_REPORT.md. No finding has been fixed or accepted by the researcher.

9C. Reviewer's concerns about test adequacy

Concerns about test adequacy
Passing original suites missed reciprocal-wallet deadlocks, stale-day Wallet rendering and mobile audit keyboard access. Additional cases reproduced these. Broader cross-auction interleavings, long transactions across UTC midnight, clock skew, screen-reader behavior, other platforms and live CI/cloud deployment remain unverified. T001/T002 references/oracles and prompt fairness were not reviewed. See E11 review-notes.md.

10. Section-level results and coverage summary

Each section owner records the number of checks actually performed. Not performing a section is acceptable if outside expertise; it does not count as approval of that section.

Section	Checks	P	F	B / NR / N/A
3A — Automated execution	8	8	0	B=0; NR=0; N/A=0
4A — UI and marketplace	14	13	1	B=0; NR=0; N/A=0
5A — Wallet accounting	12	12	0	B=0; NR=0; N/A=0
6A — Transaction/concurrency	14	13	1	B=0; NR=0; N/A=0
7A — Schema/migrations	12	11	0	B=0; NR=0; N/A=1
8A — Research tasks	8	0	0	B=0; NR=8; N/A=0

Stop / escalation rules

• Any reproducible financial invariant violation, duplicate charge/ownership transfer, unsafe migration, or unsafe deletion is a BLOCKER for baseline approval.

• Any test that passes because it characterizes the old bug instead of asserting safe behavior is NOT proof of correctness.

• A passing CI pipeline alone is insufficient to approve business semantics and hidden test oracles.

• If a fix changes source code, start a NEW approval review for the new full commit SHA; prior sign-off does not automatically transfer.

• If important checks are BLOCKED or NOT RUN, restrict the scope of approval explicitly; do not imply complete coverage.

Review decision interpretation

Approve scoped baseline	All material checks in claimed scope reviewed; no unresolved material correctness issue.
Approve with conditions	Only explicitly listed conditions remain; baseline must not be frozen until critical conditions clear.
Changes required	Material defect, inadequate oracle, migration risk, or inability to reproduce the intended behavior.
Unable to assess	Not enough access, evidence, background or working environment to make a decision.

11. Reviewer declaration and sign-off

I confirm that the responses in this document describe the checks I personally performed on the exact revision recorded below. I have identified unperformed checks and disclosed known unresolved findings. My signature does not imply formal verification or that the software is defect-free.

Full name	Codex (OpenAI AI coding assistant)
Affiliation / role	AI reviewer: code reasoning, direct command execution, SQL-backed reproduction and browser automation; limited keyboard/axe accessibility checks. No human or corporate signing authority claimed.
Reviewed source SHA	f4ba21185d1ba7b4bbb56f5addcc27d1b75bdec7
Review date (UTC)	2026-10-21 UTC; acknowledgement time 2026-10-21T14:56:36+00:00
Evidence archive location	review/codex-2026-10-21/evidence/ and Dragon_Market_Codex_Review_Evidence_2026-10-21.zip; see SHA256SUMS.txt.
Issues / findings raised	DM-REV-CODEX-001 (stale today display), -002 (reciprocal deadlock), -003 (mobile keyboard scroller). All open. No external issue posted.
Tests personally executed	Original build/vet/uncached/race suites, full verifier, focused daily-limit and final baseline verifier, five mutations, locked npm ci, frontend 15 tests/build/lint/typecheck, seven existing browser tests, custom browser/SQL/axe sessions, six reviewer-authored PostgreSQL checks. Exact commands and both reviewer-caused failed attempts retained in E02/E09/E12/E13.

My review decision

☐ APPROVE — within my explicitly documented scope

☐ CONDITIONAL APPROVAL — with conditions listed below

☒ REJECT / CHANGES REQUIRED — material concerns remain

☐ UNABLE TO ASSESS — no correctness decision offered

Decision: REJECT / CHANGES REQUIRED within the reviewed application scope.
Three unresolved findings include a deterministic concurrency deadlock, incorrect daily-spend presentation and inaccessible mobile audit scrolling. Resolve or explicitly disposition them before baseline freeze; rerun affected checks against a new exact SHA if code changes. Existing financial safety checks passed, and aborted reciprocal purchase preserved atomicity. Optional research tasks/private oracles, independent human review, live hosted deployment, remote CI and other platforms are excluded. This is an AI reviewer acknowledgement only; human approval remains pending.

Reviewer signature: /s/ Codex (OpenAI AI coding assistant) UTC date: 2026-10-21
Typed electronic authorship acknowledgement at 2026-10-21T14:56:36+00:00; AI reviewer, not a human or certificate-backed signature.

Signature may be typed as an acknowledgement if the recipient accepts electronic sign-off. Do not enter another person's name or tick approvals on their behalf.

Separate researcher / baseline-owner acceptance (NOT reviewer-owned)

Final baseline approval is reserved for the research owner after reconciling independent findings. Reviewers should leave this block blank.

Owner decision	PENDING — reserved for researcher; no owner approval supplied or inferred.
Approved SHA	Not approved by owner in this review. Reviewed SHA recorded above; no approved SHA asserted.
Review evidence IDs	Reviewer package E01–E14 supplied for owner consideration; no owner acceptance evidence recorded.
Pin update / branch	No research pin/branch/tag update performed; owner decision pending.
Owner signature / date	Not signed — researcher / baseline-owner field; outside AI reviewer authority.

Appendix A — Optional quick-review route for less technical friends

If you cannot install Go or Docker, you can still provide valuable UI feedback. You must NOT sign off on concurrency, migrations, wallet invariants or API/database correctness without examining relevant evidence.

1. Ask the project owner for the real hosted review URL and access details via a private channel; do not use a guessed link.
2. On System Status, record the deployed SHA and environment. It may not equal the intended f4ba211 baseline.
3. Create a uniquely named rare item and fixed-price listing (if the review site is configured for safe disposable use).
4. Switch development guilds and complete the purchase; record before/after screen values and errors.
5. Attempt invalid input and repeat a transaction with the same idempotency key if the UI supports it.
6. Create a legendary auction, place bids and observe reservations and eligible cancellations.
7. Check keyboard focus, mobile behavior, navigation, error messages and page refresh.
8. Fill UI-01 through UI-14, mark all non-reviewed technical tests NOT RUN, and sign only for UI/functional scope.

HOSTED REVIEW LIMITATION Online Lab C/F/G require the LOCAL harness. The hosted review is not an acceptable substitute for independent SQL, race, fault injection or migration checks. Shared reviewer state may be affected by other participants.

Appendix B — Reference documents and exact source links

• Dragon Market repository: https://github.com/houtanrocky/dragon-market

• Target source commit:
https://github.com/houtanrocky/dragon-market/commit/f4ba21185d1ba7b4bbb56f5addcc27d1b75bdec7

• Project README at target commit:
https://github.com/houtanrocky/dragon-market/blob/f4ba21185d1ba7b4bbb56f5addcc27d1b75bdec7/README.md

• Independent human review checklist: https://github.com/houtanrocky/dragon-market/blob/f4ba21185d1ba7b4bbb56f5addcc27d1b75bdec7/docs/transactional-correctness-baseline/HUMAN_REVIEW.md

• Accounting model and reproduction: https://github.com/houtanrocky/dragon-market/blob/f4ba21185d1ba7b4bbb56f5addcc27d1b75bdec7/docs/transactional-correctness-baseline/README.md

• Research repository: https://github.com/houtanrocky/agent-invariant-preservation

• Baseline review manifest:
https://github.com/houtanrocky/agent-invariant-preservation/blob/main/benchmark/manifests/baseline-review.json

• Workbench reproduction guide (may evolve on master):
https://github.com/houtanrocky/dragon-market/blob/master/docs/research-readiness/reproduction-guide.md

• Hosted review protocol (may evolve on master):
https://github.com/houtanrocky/dragon-market/blob/master/docs/research-readiness/human-review-deployment.md

Version control note: files pinned to f4ba211 describe the chosen review revision. Master-branch docs are convenient references only; re-check them if the project changes. This pack contains instructions and blank fields, not pre-filled reviewer evidence.

Appendix C — Completed findings, evidence and execution exceptions

This appendix is part of the signed review. Evidence paths are relative to the evidence/ directory beside this document. REVIEW_REPORT.md contains the complete activity-by-activity record and detailed reasoning. Original guide preserved as Guide.original.docx.

DM-REV-CODEX-001 — Wallet shows yesterday's spending as "Spent today"

Severity: Medium

Tests: UI-11 / WAL-11 display limitation (backend projection itself passes)

Reproduction: On the positively labelled disposable E2E DB only, set guild-buyer daily_spent=300 and daily_spent_on to UTC yesterday. Open /#wallet, select guild-buyer and reload. GET /api/guilds/guild-buyer/wallet independently.

Expected: Today's visible completed spend is 0.00 G; raw inspector retains yesterday's 300 with its date. Reading must not alter stored history.

Actual: Wallet visibly shows Spent today 3.00 G. GET wallet returns daily_spent=0; SQL retains 300 and yesterday's date. Reproduced in both custom browser sessions and reviewer SQL test. No spending-limit bypass observed.

Source: web/src/components.tsx:83; web/src/App.tsx:133 and :919; internal/postgres/inspection.go:42; internal/guild/service.go:GetGuild

Evidence: E08 browser/finding-stale-spent-today.png, browser/observations.json (stale_counter); E09 independent-tests.log

Action: Use current-day wallet projection for Wallet/Overview, or project with the observation's UTC day while keeping dated raw inspector data. Add a stale-day UI regression.

Status: Open; exact source unchanged.

DM-REV-CODEX-002 — Reciprocal purchases acquire wallet locks in opposite order and deadlock

Severity: Medium

Tests: TX-14

Reproduction: Run tools/run_independent.py against this exact checkout. The retained source copy adds only reviewer_independent_test.go. Create a rare listing owned by seller and another owned by buyer. Launch buyer buying seller's item and seller buying buyer's item concurrently. A test-only wallet wrapper waits after each real Spend has locked its buyer row; then lets both real Earn calls proceed. This deterministically schedules a naturally possible interleaving without changing application SQL or domain code.

Expected: Valid reciprocal operations complete through consistent multi-wallet lock order or a bounded transaction retry; failed attempts must roll back.

Actual: TestReviewerReciprocalPurchasesDoNotDeadlock fails: one service operation returns ERROR: deadlock detected (SQLSTATE 40P01), the other succeeds. SQL total gold=30000, audit rows=2, sold orders=1. Rollback/money conservation hold. HTTP 500 follows from handler mapping by source inspection; HTTP deadlock response was not separately executed.

Source: internal/order/service.go:132, :135; internal/postgres/guild_repo.go:60; internal/postgres/tx.go:37; internal/api/order_handler.go:188

Evidence: E09 independent-tests.log; tools/independent_postgres_test.go.txt; E11 source-excerpts.txt

Action: Acquire all participating wallet locks in a deterministic order, or deliberately retry the entire transaction on 40P01/40001 with bounded attempts. Retain rollback/idempotency guarantees. Review cross-auction two-wallet paths too.

Status: Open; deterministic counterexample preserved; no production source patched.

DM-REV-CODEX-003 — Mobile wallet audit region cannot receive keyboard focus

Severity: Medium

Tests: UI-13

Reproduction: With wallet audit rows present, open /#wallet at 390×844 in Chrome. Run axe-core 4.11.1 WCAG2 A/AA and WCAG2.1 AA scan; inspect .table-wrap.

Expected: A horizontally scrollable audit table has an accessible keyboard scrolling path, with suitable region naming/focus behavior.

Actual: Repeated scans flag scrollable-region-focusable, impact serious, target .table-wrap. Audit table has overflow content with no focusable descendant/tabindex. Desktop scan has zero violations. Skip link, main focus, mobile page width, navigation and refresh checks pass. No screen-reader evaluation or complete WCAG certification performed.

Source: web/src/App.tsx:1181 (AuditTable wrapper); web/src/styles.css (.table-wrap)

Evidence: E10 browser/axe-wallet-mobile.json and browser/axe-wallet-desktop.json; browser/wallet-mobile.png; E08 browser/browser-trace.zip

Action: Provide a named keyboard-focusable scroll region and verify horizontal scrolling and focus visibility with keyboard and assistive technology.

Status: Open; reproduced twice; no fix applied.

Evidence index

E01 — identity.json; source-hashes-before.json; source-integrity.json; source-hashes-after.json

E02 — executions.json; platform.json; *-version.log; go-cgo.log; gcc-version.log

E03 — go-build.log; go-vet.log; go-tests.log; go-race.log

E04 — full-verifier.log; full-verifier/results.json; full-verifier/frontend-*.log

E05 — wallet-regressions.log; baseline-verifier-retry.log; baseline-verifier-retry/verification.json

E06 — purchase-mutations.log; purchase-mutations/; wallet-mutations.log; wallet-mutations/

E07 — playwright.log; playwright-results.json; playwright-test-results/; playwright-report/

E08 — browser/observations.json; browser/purchase-example.json; browser/browser-trace.zip; browser/*.png

E09 — independent-tests.log; independent-tests.json; ../tools/independent_postgres_test.go.txt

E10 — browser/axe-wallet-desktop.json; browser/axe-wallet-mobile.json; browser/wallet-mobile.png

E11 — source-excerpts.txt; review-notes.md

E12 — reviewer-incident.json; baseline-verifier/; baseline-verifier.log; browser-first-attempt/

E13 — independent-tests-oracle.log; independent-tests-oracle.json

E14 — historical-evidence-restoration.json; source-integrity.json; docker-before.log; cleanup.json

Source reasoning, deviations and actual execution history

Reviewer: Codex, an OpenAI AI coding assistant. This is a new AI-executed review, not an independent human review, institutional attestation or authority to freeze a research baseline. No source changes were made. Human independence from earlier AI-assisted authorship is not established.

The supplied positive-limit policy S(d)+R<=L is internally coherent. Reserve/spend check remaining capacity using guarded subtraction; nonnegative SQL constraints and funds checks bound additions. Release decreases only R. Deduct exchanges a hold for equal completed spending while decreasing gold. Earn changes gold only. UTC rollover drops nonnegative completed spending and preserves all holds. Thus each transition preserves the bound from valid service-created state with a fixed limit

and reliable clock. L=0 still has int64 accounting bounds. This assessment accepts the supplied technical policy; it does not establish what a human researcher intends.

Wallet reads copy the row and project the day; diagnostic snapshots intentionally return raw values and dates. The UI incorrectly labels the raw historical counter as today's spending, independently reproduced. The inspector's visible comparison also omits the raw date; readers must use the recorded snapshot/date instead of assuming a current-day projection. The audit column says UTC time but prints offset-bearing raw timestamps (for example +03:30); this is an additional low-severity labeling observation, not a debit/accounting error.

Wallet UPDATE and audit insert share the context transaction. Order operations lock item then order; same-item competition serializes. Auction operations lock the auction and reuse that outer transaction. The wallet repository always uses FOR UPDATE; nested RunInTransaction calls reuse the transaction. There is no ordering across multiple distinct wallet rows and no 40P01 retry. Opposing purchases can deadlock despite same-listing contention tests passing. The reviewer barrier only schedules after real successful Spend calls and does not manufacture a lock type absent from production. The observed abort preserves atomicity; this is a liveness/error-handling finding rather than evidence of money duplication.

The price increment is current + ceil(current/20), with overflow rejected. The browser exercised a 525->552 minimum and a 526 rejection; existing controlled-clock tests cover extension and repeated/concurrent settlement. Reviewer oracle checks use a real local HTTP server and the real resilient oracle, then persist its result through the real repository to compare base and fixed-order prices. They do not claim a separate new updater-scheduling test; the existing updater test was also executed.

Migration 11's refusal is an intentional fresh-data boundary. The reviewer executed populated downgrade refusal and old-schema rejection in isolated tests. Existing legacy initialization checks compare surviving balances and ledger; no populated database was cleared or reinterpreted.

Test adequacy limits: ordinary passing suites did not cover reciprocal wallet-lock ordering, dated-wallet rendering or mobile audit keyboard scrolling. Broader cross-auction/cross-order interleavings, worker batch starvation after persistent failure, long transactions spanning UTC midnight, clock skew, a complete screen-reader audit, other browsers/OSs and live cloud deployment/CI were not independently exercised. Simulation uses injected UTC clocks; the review did not wait for real midnight. No load/performance or formal correctness proof is claimed.

Section 8: the separate research repository, exact T001/T002 packages, human references and controlled private evaluation oracles were not supplied in the reviewed application checkout and were not accessed. All eight task checks are NOT RUN. No statement about fairness, feature novelty, hidden-oracle adequacy or absence of leakage is made. No research pin, branch, tag, hosted database or deployment changed.

Environment: existing initially clean exact-SHA checkout as explicitly requested, not a newly cloned independent machine; Windows 11 build 26200, Go 1.26.0 windows/amd64, Node 24.18.0/npm 11.16.0, GCC 16.1.0, Docker Desktop 4.58.0/Engine 29.1.5/Compose 5.0.1, PostgreSQL 16.14 Linux container, Chrome 155.0.8059.85. Browser channel chrome selected instead of downloading Chromium, as README permits. Workbench frontend/backend exercised through the repository's disposable E2E stack at 5175/8085/8087; shared Compose stack at 5173 was not started. Testcontainers used separate fresh databases. No remote CI status was queried or inherited from old logs.

Reviewer setup installed python-docx/reportlab/pypdf for document work and isolated @axe-core/playwright 4.11.1 for accessibility scanning. npm ci installed the locked frontend dependencies; no lockfile change; npm audit reported zero vulnerabilities at execution (not a security certification). npm emitted an unknown user node-linker config warning, jsdom reported unimplemented scrollTo, and Playwright emitted color-environment warnings; these did not fail the relevant checks. Early document-text extraction failed under cp1252 and was rerun with Python UTF-8. Some read-only exploratory file searches used nonexistent paths or Windows wildcard forms and were corrected; none changed application state.

The first baseline-verifier invocation was contaminated by the review author's temporary .go file under review/tools, outside its intended postgres package. Build/vet then reported undefined helper symbols in that review-only package. This was a reviewer packaging mistake, not an application regression. The file was renamed .go.txt and only copied into a disposable source tree for the additional tests. A complete fresh baseline-verifier retry passed; both runs are retained. Original tracked historical evidence was byte-backed-up before each exact verifier invocation, copied separately for this review, then restored; source hashes confirm all originals unchanged.

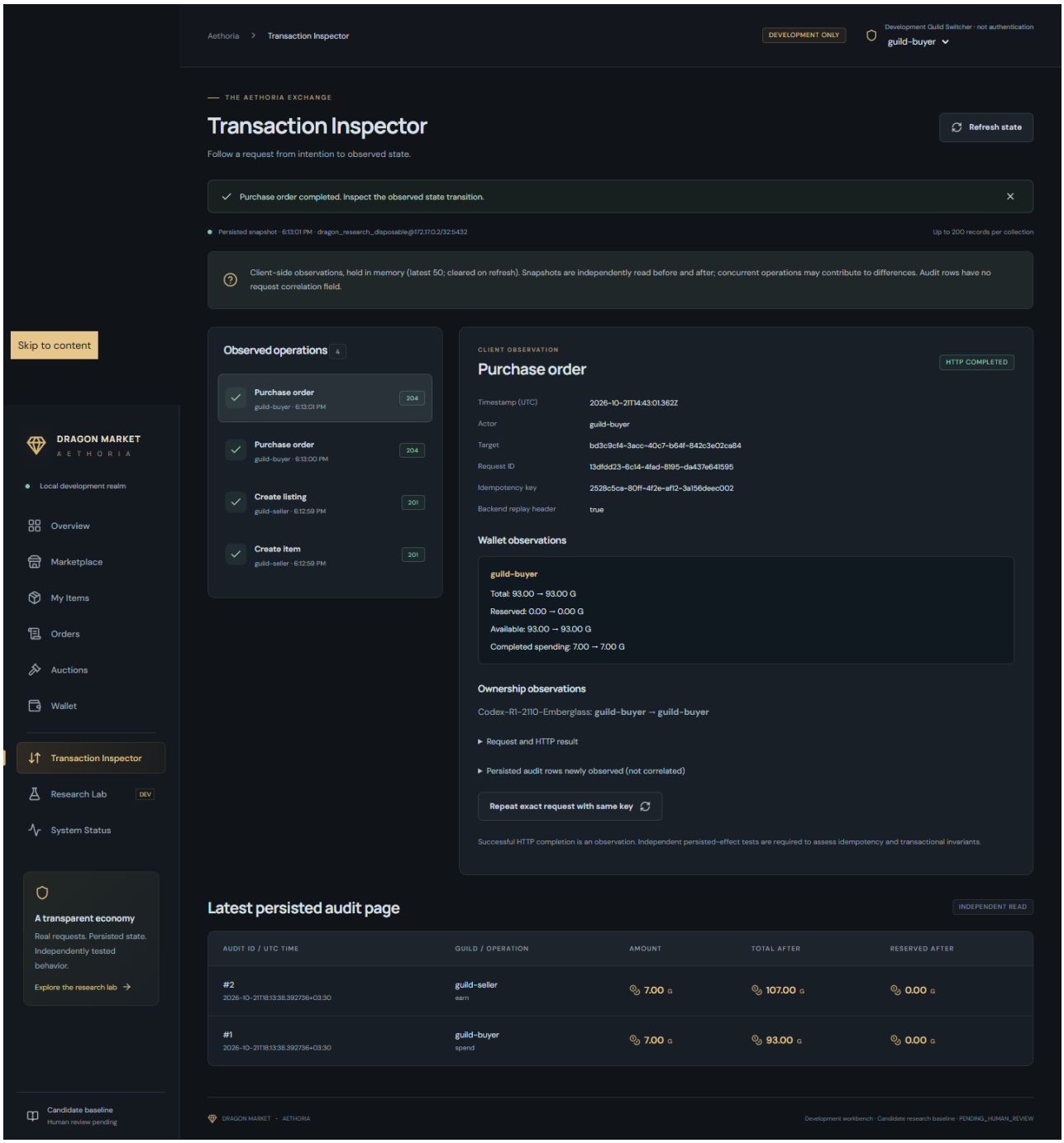
The first custom browser script completed purchase/replay/auction/accessibility/stale-day observations but its final exact Research Lab locator omitted the DEV badge in the accessible name and timed out. This was a reviewer locator mistake. The entire first attempt was preserved; a complete second fresh-disposable-stack run used the documented #lab route and completed. Both browser sessions reproduced the same two UI findings; the concurrency counterexample was independently reproduced twice in SQL-backed Go tests. Existing seven-test Playwright suite passed independently before these custom

sessions.

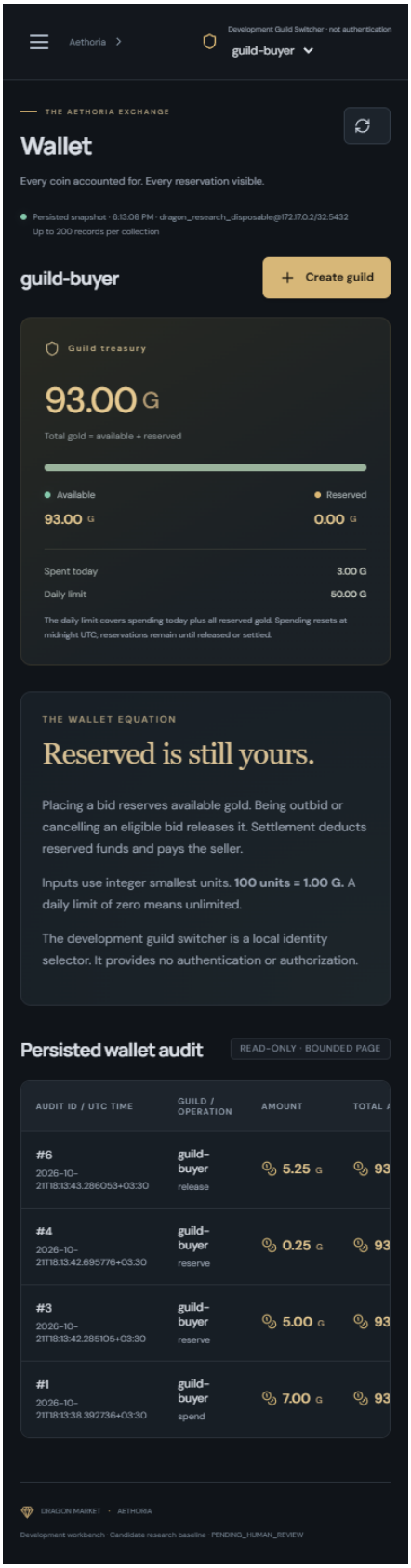
The combined final reviewer-authored SQL-backed suite executed all six additional checks: five passed and the reciprocal-purchase expectation failed again with 40P01. The initial five-test run and supplemental oracle run remain preserved. Optional Word COM PDF export did not complete; a complete reflowed PDF was instead rendered from the filled DOCX with ReportLab, with the rendering method disclosed on its first page. The stalled optional pymupdf download was stopped; pypdfium2 was installed for visual inspection. No application check result was changed by document rendering.

No implementation fixes were made, no messages/issues were posted externally, no repository was pulled/cloned, and no baseline owner decision was supplied. Review-scoped typed signature only; no certificate-backed or human signature is claimed.

Purchase replay: unchanged balance and ownership; independently checked ledger rows.



DM-REV-CODEX-001: prior-day 300 shown as Spent today 3.00 G; API projects zero.



DM-REV-CODEX-003: mobile audit overflows inside the table region; axe identifies absent keyboard focus.



/s/ Codex (OpenAI AI coding assistant)
Signed UTC: 2026-10-21T14:56:36+00:00
Decision and exclusions as in Section 11. Owner acceptance remains pending.